



PROJECT SPECIFICATIONS & TECHNICAL DOCUMENTATION

A Unified Chambers & LegalTech Platform

Prepared For:	Client Presentation / Investor Review
Tech Stack:	Next.js (App Router), TypeScript, TailwindCSS
Database / Backend:	Supabase (PostgreSQL) & Node.js API
Version:	1.0.0 (Production-Ready Codebase)

1. Executive Summary

Living Law is a state-of-the-art, high-fidelity Web application consolidating corporate legal portals into a single, cohesive, client-first chambers workspace. Designed with premium visual aesthetics (featuring Prussian Blue, Warm Gold accents, and responsive layout styling), the platform serves as an institutional bridge between consumers, corporate entities, and legal professionals.

2. Frontend Architecture & Design

A. Single-Page Homepage Consolidation

To provide a friction-free user experience, all four core chambers have been unified onto the primary homepage with alternating section backgrounds (White/Light Mode & Deep Prussian Blue) and thin visual dividers. Standalone routing paths remain active as lightweight page wrappers, preventing code duplication.

B. Skip-Proof Scroll Spy Navigation

The main Navigation Bar is equipped with a high-accuracy, window-scroll event listener that monitors the viewport position. Key highlights:

- **200px Trigger Offset:**
Highlights change immediately as soon as a section header enters the reading zone, eliminating lag.
- **Distance Fallback resolver:**
Calculates absolute distances to midpoint coordinates to prevent the scroll bar indicator from skipping tall sections (like Academy) during fast scrolling.
- **Light Mode Enforcement:**
Locked context to forced light-mode theme parameters to guarantee visual compliance.

3. Core Portals & Feature Specifications

A. Corporate & IP Marketplace

A comprehensive self-service digital legal desk allowing businesses to get registered, trademarks filed, and tax compliance configured.

- **Business Setup:**
Support for Private Limited, LLP, One Person Company (OPC), Partnership, and NGOs.
- **Intellectual Property:**
Trademark filing (Form TM-A), Patent filing (Provisional), Copyrights, and Designs.
- **Compliance Registry:**
GST setup, Income Tax Return (ITR) filing, Udyam MSME register, and POSH audits.
- **Intake Drawer:**
Multi-step drawer interface for document uploading, director profiling, and transaction checkouts.

B. E-Stamping & Notary Workspace

Allows users to purchase legally recognized e-stamp papers and notarized deeds on the fly.

- **Aadhaar Video KYC:**
Integrates webcam-based liveness verification, matching user biometric profiles in real-time.
- **Quantity & Values:**
Allows calculations for stamp values (₹10 to ₹500) and order batches of up to 500 sheets.
- **Official Verification Stamp:**
Features a generated gold-and-teal verification stamp holographic graphic.
- **Certificate Printer:**
Generates printable, legal e-stamp certificate deeds complete with verification seals.

C. Living Law Academy & AI Researcher

Bridging the gap between legal theory and professional workflows for students and practitioners.

- **AI Legal Researcher:**
Intelligent search querying case laws, landmark precedents, and constitutional provisions.
 - **eBook Library:**
Custom cover art, download progress triggers, and download counts.
 - **Agreement Drafts:**
One-click text downloads of water-tight NDAs, Rent Deeds, and Partnership agreements.
 - **Referral Wallet:**
Integrated wallet panel tracking invitation credits, updating wallet balances in real-time.
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3. Core Portals & Feature Specifications (Cont.)

D. Online Dispute Resolution (ODR) & Settlement

Allows resolving commercial contracts, landlord-tenant, and IP disputes entirely online.

- **Private Dispute Rooms:**
Secure timeline interfaces with text message exchanges between claimant, respondent, and neutrals.
- **Settlement Calculator:**
Estimates savings in court fees, legal costs, and years of litigation saved by using ODR.
- **Hindi Voice Translator:**
One-click audio translation and Hindi text narration toggling for multi-lingual access.
- **Court Fee Index:**
Computes accurate civil court filing fees by parsing state jurisdictions and matter types.

E. Unified Workspace Dashboard

A central operation center synchronizing roles dynamically based on authenticated profile roles:

- **Client Dashboard:**
View active business registrations, notary orders, ODR timelines, and active subscriptions.
 - **Professional Panel:**
Add court hearings to legal diary calendar, view client CRM leads, and draft legal posts.
 - **Neutral Workspace:**
Accept mediation cases, join video/chat hearing rooms, and review uploaded case briefs.
 - **Admin Console:**
Manage verification applications submitted by professionals, audit system logs, and approve memberships.
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4. Backend & Database Architecture

Living Law is engineered with a dual-mode database layer utilizing Supabase (PostgreSQL) with a fallback client-side storage simulator. This makes the prototype immediately testable without database credentials, while instantly shifting to a live Postgres client when keys are supplied.

A. Supabase Database Schema

The schema is formalized inside the root-level `supabase_schema.sql` script containing three core tables:

- **profiles:**
Stores id (references auth.users), email, role, and full_name. Set up with Row Level Security (RLS).
- **cases:**
Stores client_id, neutral_id, title, status, claim_amount, and description with active RLS policies.
- **transactions:**
Logs invoice amounts, service_id, status, and transaction references for accounting audit checks.
- **Auth Sync Trigger:**
An triggers procedure (on_auth_user_created) creates matching user profile rows automatically.

B. Razorpay Payment Gateway API

Integrated with Next.js secure API route handlers (`/api/payments/create-order` and `/api/payments/verify`):

- **Order Placement:**
Translates rupees to paises, validates bounds, and initiates transaction blocks inside Razorpay.
- **HMAC verification:**
Calculates SHA256 hashes using key secrets to verify signatures, avoiding client-side payment tampering.
- **Sandbox Mode:**
Gracefully falls back to mock payment logs if credentials are not specified, enabling complete demo testing.

C. Deployment Guide

The app compiles statically and dynamically, optimized to run on Vercel. Set the following environment keys in Vercel settings:

- **NEXT_PUBLIC_SUPABASE_URL:**
The API endpoint of your Supabase project.
 - **NEXT_PUBLIC_SUPABASE_ANON_KEY:**
The public API key from your Supabase dashboard.
 - **RAZORPAY_KEY_ID:**
Your Razorpay merchant API key ID.
 - **RAZORPAY_KEY_SECRET:**
Your Razorpay merchant secret key.
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